

KULLIYAH OF ENGINEERING
DEPARTMENT OF MECHATRONICS ENGINEERING

PROJECT REPORT :
INDUSTRIAL BATCH OBJECT COUNTER



TAWHIDIC EPISTEMOLOGY **UMMATIC EXCELLENCE**
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INDUSTRIAL INSTRUMENTATION

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TABLE OF CONTENT

1.0 INTRODUCTION	
1.1 PROBLEM STATEMENT.....	3
1.2 APPLICATION OVERVIEW.....	4
1.3 TECHNOLOGY GAP.....	4
 2. SENSOR SELECTION AND SYSTEM DESIGN	
2.1 SENSOR SELECTION RATIONALE.....	5
2.2 SYSTEM ARCHITECTURE.....	6
 3. SIMULATION AND INTERFACE DESIGN	
3.1 SIMULATION DEVELOPMENT.....	7
3.2 LABVIEW INTERFACE DESIGN.....	7
3.3 PROGRAM LOGIC.....	10
 4. REAL SENSOR IMPLEMENTATION AND TESTING.....	11
4.1 HARDWARE SETUP.....	11
4.2 LABVIEW INTEGRATION.....	12
4.3 TESTING AND OBSERVATION.....	12
 5. REFLECTION	
5.1 TECHNICAL AND PROFESSIONAL SKILL LEARNED.....	13
5.2 CHALLENGES FACED AND PROBLEM SOLVING METHOD.....	13
 6.0 CONCLUSION AND RECOMMENDATIONS	
6.1 CONCLUSION.....	14
6.2 LIMITATION.....	14
6.3 SUGGESTION FOR IMPROVEMENT.....	14

1.0 INTRODUCTION

1.1 PROBLEM STATEMENT

- **Manual Counting Error:** Operators suffer cognitive fatigue during repetitive tasks, leading to inaccurate counts and inconsistent batch sorting.
- **Throughput Bottlenecks:** Manual logging cannot match high-velocity production packaging speeds, necessitating automated real-time tracking frameworks.

1.2 APPLICATION OVERVIEW

- **Core Processing Unit:** An ESP32 microcontroller processes edge-triggered digital pulses via non-contact optical beam interruption.
- **Alert System Peripherals:** Microcontroller drives local alerts using an RGB LED indicator and a physical piezo buzzer for critical status changes.
- **HMI Remote Supervision:** Count data strings are structured and streamed to update a remote graphical LabVIEW user monitoring interface.

1.3 TECHNOLOGY GAP

- **Traditional Approach Limitations:** Mechanical limit switches or tripwires suffer rapid mechanical wear and risk damaging fragile products.
- **Proposed Optical Improvement:** Non-contact optical sensing eliminates mechanical friction, drops maintenance overhead, and processes data with minimal latency.

2. SENSOR SELECTION AND SYSTEM DESIGN

2.1 SENSOR SELECTION RATIONALE

To implement a cost-effective design pathway, the prototyping phase distinguishes between the target industrial environment and the laboratory simulation constraints.

SENSOR	FUNCTION	REASON TO PICK	ADVANTAGE
INFRARED SENSOR (IR)	Object detection via beam interruption	Chosen for simulation phase; matches industrial logic and provides high cost optimization .	Extremely low cost, zero mechanical wear, simple 3-wire digital interface.
ULTRASONIC	Distant measurement using sound wave	Not selected because distance variations can cause erratic count triggers.	Good for distance tracking, but slow response time for fast-moving items.

LIMIT SWITCH	Physical contact counter	Not selected due to mechanical degradation and potential to damage products	Very low cost and simple logic, but suffers from physical wear and tear.
PHOTOELECTRIC SENSOR	Heavy-duty industrial object detection	Identified as the target sensor for actual industrial deployment due to high environmental resilience.	Long sensing range and high dust/glare resistance, but expensive for lab simulation.

Working Principle & Cost Optimization: Both the IR sensor and the Industrial Photoelectric sensor operate on the exact same fundamental principle of optical beam interruption. When an object passes through, the infrared beam is broken, causing the sensor's digital output to change state. Since their software logic profiles and electrical signals are identical, the low-cost IR sensor was chosen for the laboratory simulation phase to minimize hardware costs while fully validating the industrial-grade control code.

2.2 SYSTEM ARCHITECTURE

Data Input Layer: Routes digital state signals from the optical IR sensor and a hardware reset button directly into ESP32 GPIO channels.

Microcontroller Processing Hub: The ESP32 tracks edge triggers, calculates the total count registry, runs software debouncing, and controls alerts.

Output Peripherals: Drives an RGB LED for visual states, a piezo buzzer for hardware alarms, and outputs UART serial string data packages.

Host Monitor Communication: Streams formatted tracking logs via a USB-UART link at 115200 baud to feed the active LabVIEW HMI block diagrams.



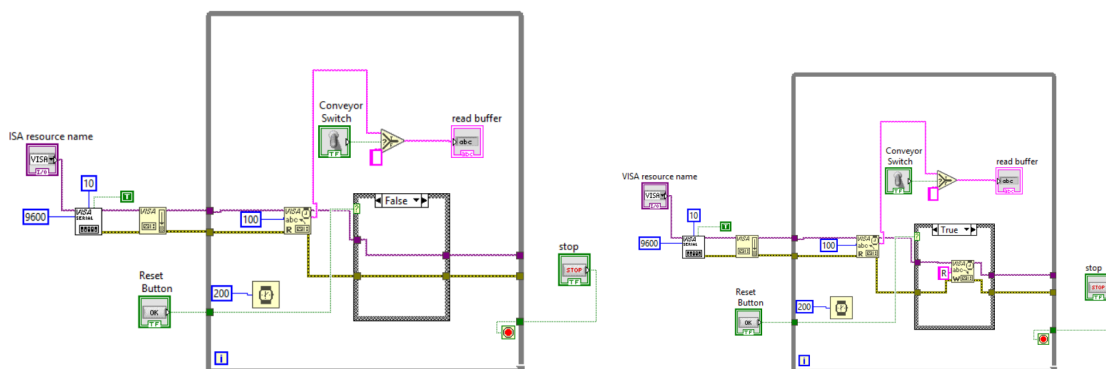
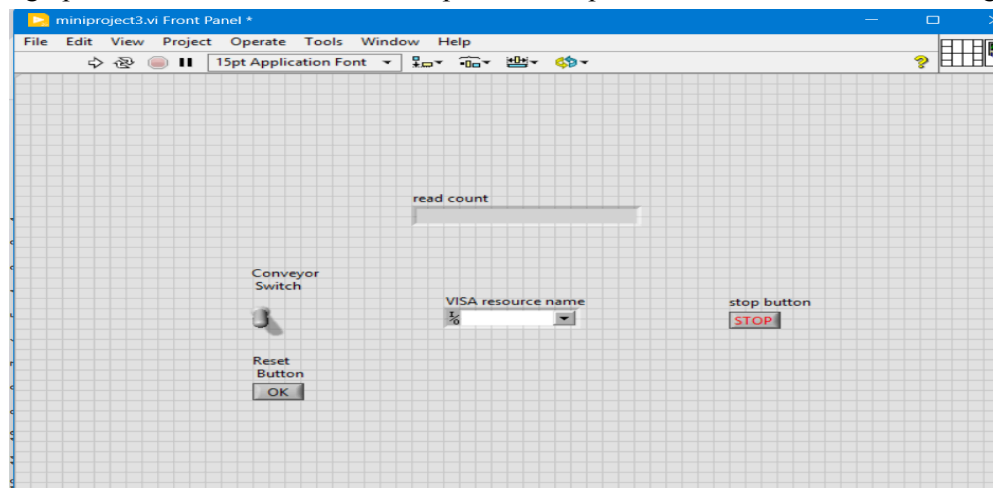
3. SIMULATION AND INTERFACE DESIGN

3.1 SIMULATION DEVELOPMENT

- Firmware Testing: The simulation verified the digital polling script and system pin layouts before moving to permanent physical assembly.
- Noise Elimination: Physical conveyor items often produce signal bounces due to line vibrations or uneven product contours.
- Software Debouncing: A temporal lockout filter (`delay(150);`) was programmed into the main execution loop to pause reading for 150ms after a trigger, ensuring each physical item records exactly one count.

3.2 LABVIEW INTERFACE DESIGN

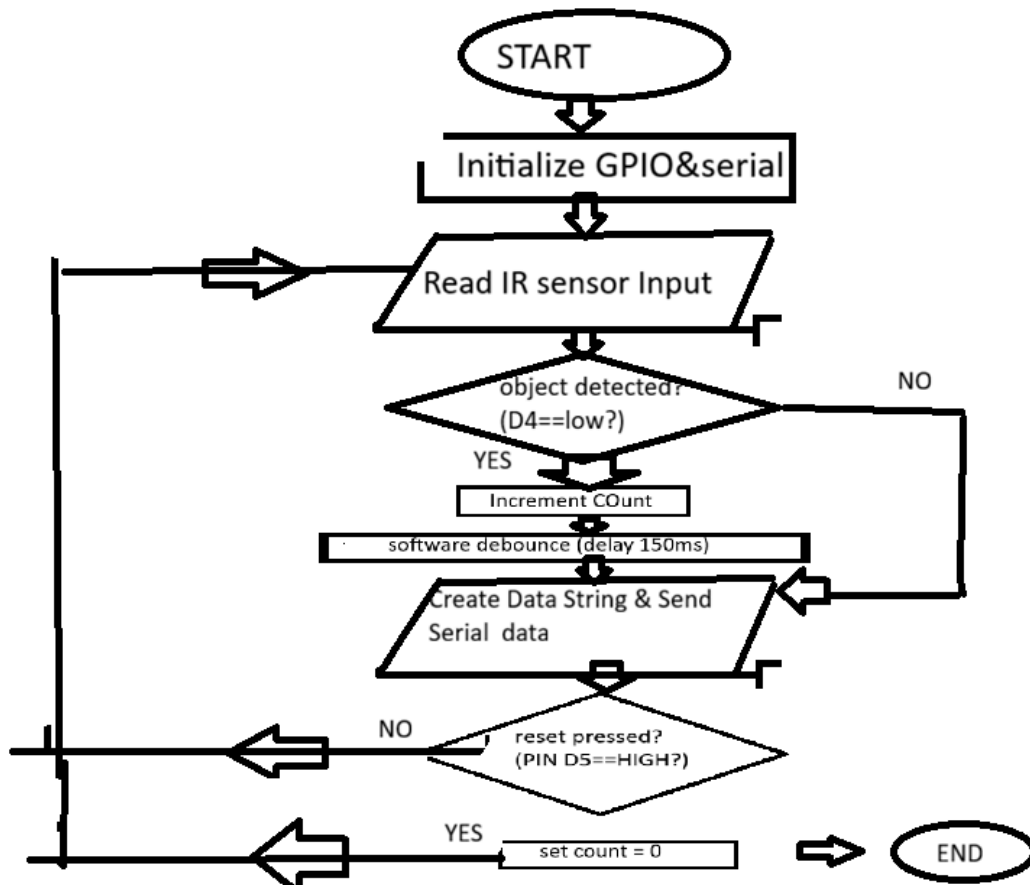
HMI Dashboard Layout: The front panel displays an automated numerical indicator for total counts, a graphical trend waveform chart for production speed, and virtual boolean status lights.



- **Serial Resource Integration:** The block diagram leverages the **NI-VISA Resource Library** to configure port connection
- **Live Buffer Tracking:** A running loop executes a VISA READ command block to constantly pull, extract, and parse incoming data strings for real-time visualization

3.3 PROGRAM LOGIC

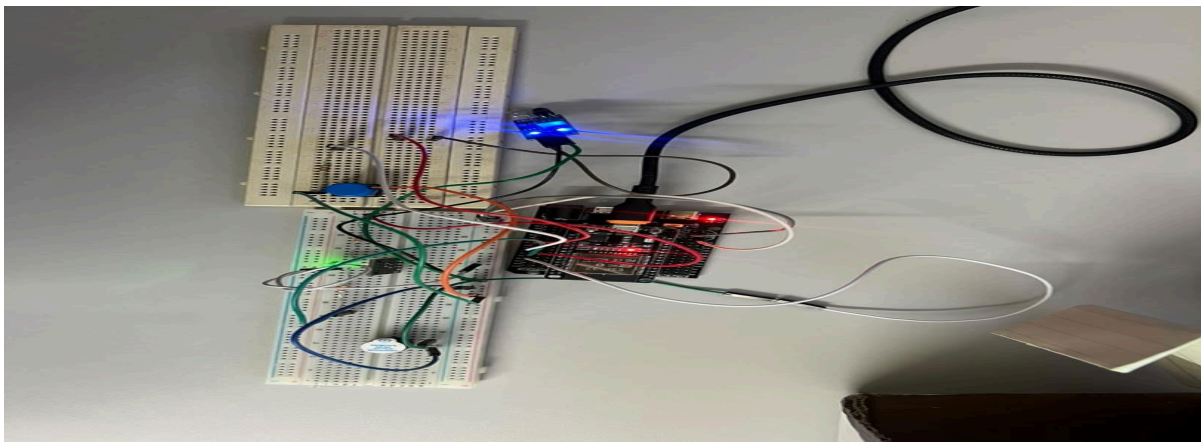
- **Monitoring & Detection:** System boots, loops continuously, and monitors the digital sensor channel for state drops to logic **LOW**.
- **Increment & Debounce:** Valid object triggers execute the register counter update function (**count++**) and apply the 150ms safety lockout delay.
- **Data Packaging & Streaming:** Packages data values into standard strings, streaming them via USB-UART to update the host computer dashboard.
- **Button Query & Reset:** System checks the reset pin state; a physical button press forces the registry back to zero to clear ongoing cycles.



4. REAL SENSOR IMPLEMENTATION AND TESTING

4.1 HARDWARE SETUP

The hardware prototype consists of an ESP32 development board, an IR obstacle sensor, a push button, a piezo buzzer, and an RGB LED indicator integrated on a standard breadboard using jumper wires. The ESP32 serves as the primary DAQ and controller unit, powered directly through a host USB connection. The hardware sub-circuits were fully validated for electrical continuity and logic response prior to establishing communication links with the host computer software.



Components used:

1. Esp 32
2. IR Sensor
3. Breadboard
4. Jumper wires
5. Buzzer
6. RGB light

4.2 LABVIEW INTEGRATION

LabVIEW functions as the central graphical user interface (HMI) to handle real-time monitoring and data visualization. A stable hardware-to-software serial bridge was established over a physical USB cable link. The designated host COM port is mapped and configured via the **VISA Configure Serial Port** node. The incoming data stream package is captured through a continuous loop running a **VISA Read** block, which strips, decodes, and routes the numerical tracking data directly onto the front panel displays.

4.3 TESTING AND OBSERVATION

The system was subjected to strict operational testing across distinct conditions to verify execution accuracy and system stability.

Test No	Condition	Expected result	RGB light	Actual Result
1	Object detected	Count increases	Turn green	success
2	No object	Count remains	Turn red	success
3	Push button press	Count Reset	Turn off	success

5. REFLECTION

5.1 TECHNICAL AND PROFESSIONAL SKILL LEARNED

- **Microcontroller Configuration:** Gained practical skills in handling ESP32 GPIO mapping, DAQ setups, and sensor signal acquisition properties.
- **HMI Design Controls:** Developed code structural fluency using LabVIEW graphical programming and managing serial streams via NI-VISA blocks.

5.2 CHALLENGES FACED AND PROBLEM SOLVING METHOD

- **NI-VISA Driver Failure:** Encountered frequent, random corruption of the host computer's NI-VISA driver, which locked and rejected the COM ports.
- **Electrical Ground Noise:** Laptop battery chargers bled AC ripple noise and ground currents into the USB connection, corrupting the serial data packets.
- **Technical Resolution:** Solved by executing clean driver reinstalls, isolating the power rails, and using a 150ms software debouncing filter.

6.0 CONCLUSION AND RECOMMENDATIONS

6.1 CONCLUSION

the project achieved its core objectives by delivering a low-cost, automated alternative to traditional manual industrial counting frameworks. Despite software driver hurdles, the prototype accurately logs passing items and streams data packages over serial networks. The optical non-contact methodology minimizes human counting errors and labor reliance, showing that combining affordable microcontrollers with graphical supervisor HMI programs is highly practical for modern industrial instrumentation designs. instrumentation layouts.

6.2 LIMITATION

- **Physical Tethering Constraints:** The current architecture strictly dictates a continuous, hardwired USB cable connection to link the microcontroller to the terminal PC, creating spatial layout constraints and exposing the raw data stream to external factory floor electrical noise over long distances
- **Environmental Vulnerabilities:** The infrared optical path remains highly susceptible to ambient factory light glare and rapid dust accumulation, which could easily block the beam or lead to erratic, false count triggers in actual harsh industrial environments
- **Software Interfacing Instability:** The system's operational runtime is heavily dependent on driver stability, meaning any sudden corruption within the NI-VISA library completely freezes the real-time HMI supervisory dashboard.

6.3 SUGGESTION FOR IMPROVEMENT

To build upon the current design, several technical recommendations should be considered for future iterations of this counting system. First, the physical USB serial bridge can be replaced with wireless communication protocols, utilizing the microcontroller's internal Wi-Fi capability to seamlessly stream live throughput data over an MQTT network. Additionally, integrating a dedicated cloud data-logging service like ThingSpeak or Firebase would allow factory plant supervisors to monitor production counts remotely from any standard web browser or mobile application. Finally, upgrading the basic infrared module to a fully sealed, modulated time-of-flight laser sensor would drastically improve the system's resilience against ambient glare, dust buildup, and unexpected environmental lighting variations, making the hardware much more robust for high-speed automated sorting applications.